Glider Deployment Report: unit_1024 (March 16, 2026)

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Summary

The Ecosystem Science Division (ESD) at the Southwest Fisheries Science Center (SWFSC) deployed glider unit_1024 (1024) on March 16, 2026 off the coast of Southern California (33.23°N, -117.49°W) (Figure 1). Sensors deployed on the glider are listed in Table 1.

Figure 1 (A) displays tracklines of glider during deployment, while (B) displays the broad deployment area relative to land. The glider remained deployed for 4 days, performed 86 dives, and traveled a total of 53.05 km while diving to a maximum depth of 503.7 m. No mission aborts occurred and the glider was recovered on 2026-03-19.

Mission specific information… include mission goals, additional glider information, etc

UPDATE FOR SHADOWGRAPH DEPLOYMENTS The shadowgraph camera was programmed to take photos every 2 seconds on dives and climbs, with a XX% red flash brightness. During the deployment, the shadowgraph camera took XX photos and used XX% of the camera storage (XX.X GB). **TODO** - pull from updated netCDF.

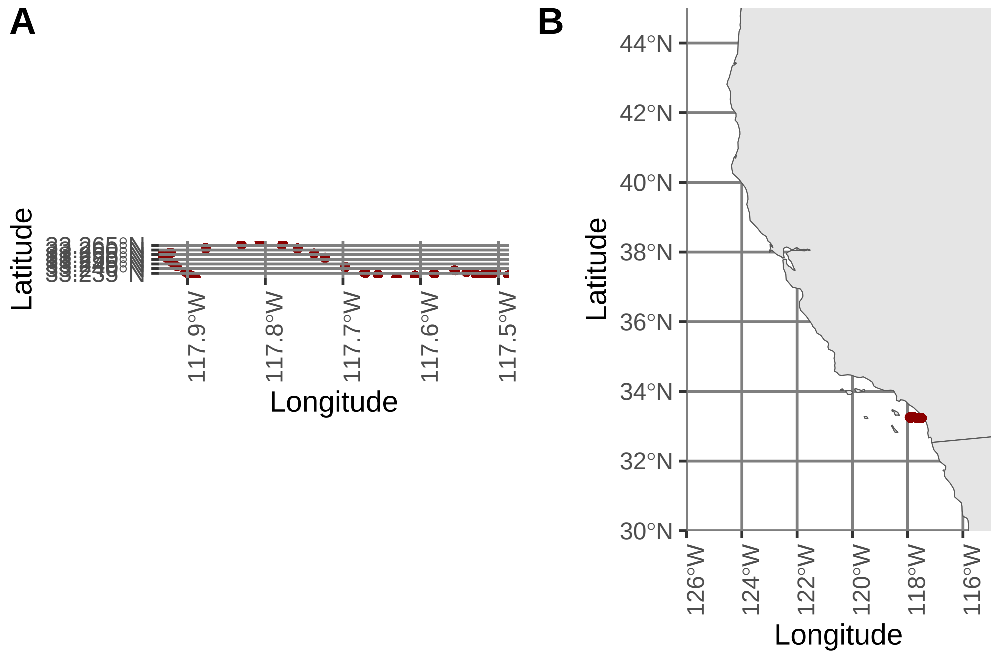


Figure 1: Glider tracklines. A displays close-up tracklines, while B displays the broad deployment area.

Table 1: Science sampling strategies for current glider deployment. Additional settings for the AZFP echosounder, the Williamson and Associates camera, the Nortek echosounder (if installed), and passive acoustic monitoring sensors (WISPR or DMON) are defined in configuration and initialization files on the glider's science computer, and are also housed on the Google Cloud Platform. All deployment files are available on request.

File Name	Sensor	State to Sample	Depth to Sample	Serial Number
sample01.ma	Sea-Bird Conductivity Temperature Depth (CTD) (SBE-41)	Diving, hovering, climbing	1000 m	9715
sample48.ma	Sea-Bird ECO Puck (backscatter and fluorescence) (FLBBCD-SLC, CDOM)	Diving, hovering, climbing	1000 m	6866
sample54.ma	AANDERAA oxygen optode (4831)	Diving, hovering, climbing	1000 m	1160
sample87.ma	Shadowgraph camera (Williamson and Associates)	Diving, hovering, climbing	300 m	SG-V3-003

Introduction

The Ecosystem Science Division at NOAA Fisheries' Southwest Fisheries Science Center monitors the living marine resources within the Southern Ocean and the California Current in order to satisfy the requirements of several legislative mandates to support conservation and management decision-making. To achieve this goal, we use autonomous underwater buoyancy-driven gliders with integrated sensors for measuring ocean conditions, plankton densities, and marine mammal distributions.

Depending on the specific deployment objective, Slocum gliders are equipped with a suite of sensors. We obtain acoustic estimates of zooplankton density (primarily Antarctic krill in the Southern Ocean) using one of two different echosounders: an Acoustic Zooplankton Fish Profiler with discrete frequencies at 67.5 and 125 kHz (AZFP, ASL, Inc) and a mini-Signature 100 wideband echosounder with continuous frequencies between 70 and 120 kHz (Nortek). We also collect ancillary oceanographic data (temperature, salinity, dissolved oxygen, chlorophyll, colored dissolved organic matter, backscatter, and photosynthetically active radiation) to characterize the marine environment. Additional sensors may include passive acoustic monitors for marine mammal detection ("Wispr", Embedded Ocean Systems; digital acoustic monitoring "DMON", Woods Hole Oceanographic Institution), "glidercams" for verifying acoustic targets (Williamson and Associates, Inc.) and shadowgraph cameras for obtaining imagery of the plankton community (Williamson and Associates, Inc.). Imagery is used to train artificial intelligence (AI) models to automate plankton identification.

Hefring OceanScout gliders are equipped with conductivity-temperature-depth sensors (CTDs; RBR, Inc.) and "Wispr" passive acoustic monitors.

Pre-Deployment Preparation and Testing

Prior to deployment, the ESD has a standard protocol for preparing and testing gliders to minimize or eliminate issues that may occur due to human error during deployment:

Slocum gliders

1. Gliders are properly ballasted (i.e., weighted) so that the density of the glider matches the density of the water in which it will be deployed. Weight and flotation configurations are documented
2. The junctions between glider sections are thoroughly cleaned, old o-rings are discarded, new o-rings are inspected for damage that may compromise their ability to form a water-tight seal, new o-rings are properly lubricated, and the glider is sealed together. All cable connections are photographed to document the "final seal" and ensure the glider was reassembled properly

3. A “Functional Checkout” is performed to ensure and document that all glider systems and science sensors are functioning properly. During the Functional Checkout, we verify the battery type installed in the glider (lithium primary or lithium rechargeable) and that the appropriate battery duration (total coulomb amp hours) is active in the glider’s autoexec.mi file
4. Two test missions are performed in the SWFSC test tank (20 m x 10 m x 10 m) to ensure the glider is performing as expected
5. Once per year, glider compasses are calibrated (the compass was not calibrated prior to this deployment)
6. Biofouling prevention measures are applied as necessary
7. Glider flight and science sampling files are prepared according to mission objectives. These objectives are identified by the Principal Investigators for each deployment. Files are uploaded to the Teledyne Webb Research Slocum Fleet Mission Control (SFMC) web interface and sent to the glider just prior to deployment over the Iridium connection
8. When gliders are shipped to their deployment location, ESD glider technicians perform a second Functional Checkout to ensure the gliders function properly after transit

OceanScout gliders

1. Gliders are configured with the standard ballast for salt water (8 weights in the nose)
2. O-rings are replaced as needed
3. Motors (variable buoyancy engine, pitch, and roll) are calibrated, either through the glider’s wifi connection prior to deployment, or by using platform.test through the command line interface
4. Routes, diving parameters, and sensor sampling strategies are established through the Hefring cloud interface and deployed to the glider

Deployment-specific testing

Deployment

unit_1024 was deployed on 2026-03-16, approximately **UPDATE - 5 km west of Mission Bay in the Pacific Ocean** (33.23°N, -117.49°W) from the ESD Zodiac R/V *Ernest II*. This glider was deployed with the sensor configuration listed above, and with batteries (coulomb amp hour total = 300). We began this deployment using the autoballast feature to maximize oil pump efficiency. Autoballast converged **UPDATE - successfully early on and remained converged for the duration of the deployment**.

Because digifins are sensitive to slight movements and produce erroneous oddities that can lead to mission aborts, the following commands were issued to the glider prior to sequencing our standard mission (1k_n.mi):

```
put u_digifin_hide_oddities_at_surface 1
```

```
put u_digifin_mask_movement_warning_at_surface 1
```

UPDATE - add deployment information on issues with test/initial dives, aborts, errors, etc.

This glider performed well for the entire 4-day mission. The glider used 24.03 amp hours over 4 days, or 8.01% of its battery capacity.

UPDATE - add sensor sampling information Throughout the deployment, pilots systematically turned all science sensors (except the PAM) on and off to evaluate whether noise from these sensors affected the quality of PAM data. The PAM remained on for the duration of the deployment. Other sensors were cycled on and off according to Table 2. When sensors were on, they sampled only on dives.

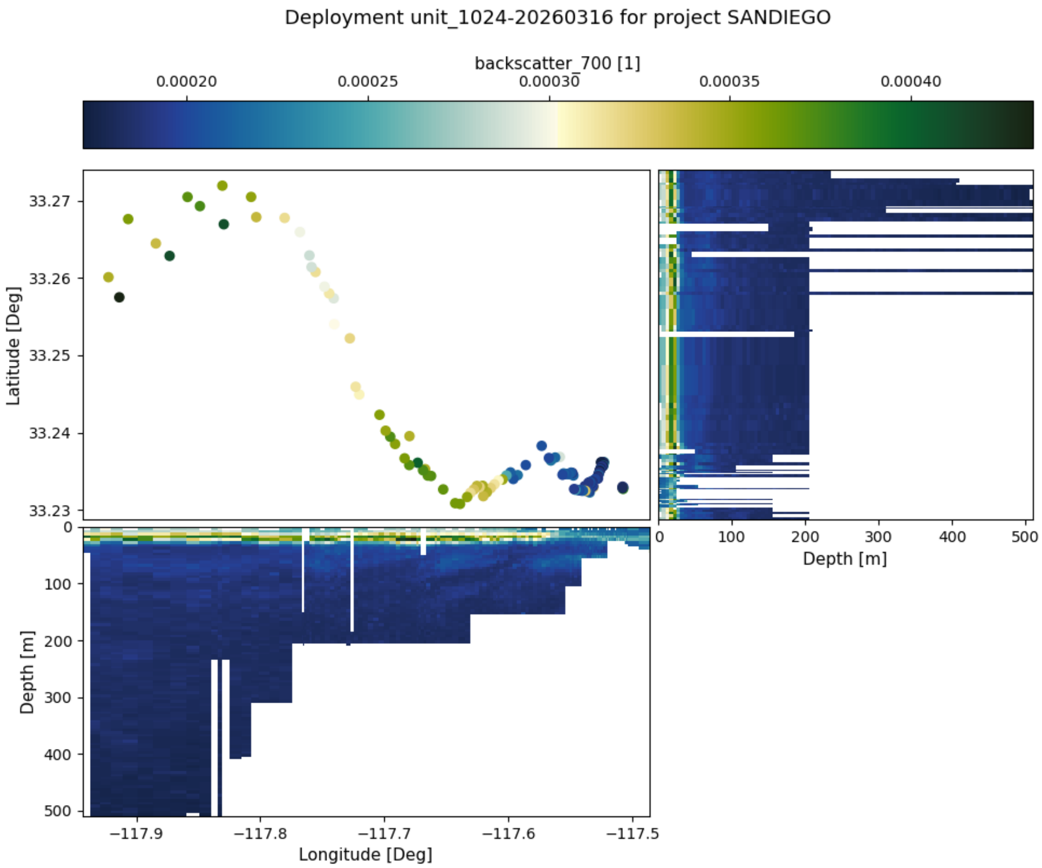
Post-deployment actions

Once the glider was back in the laboratory, pilots downloaded data via ___ and uploaded to ___ **UPDATE**

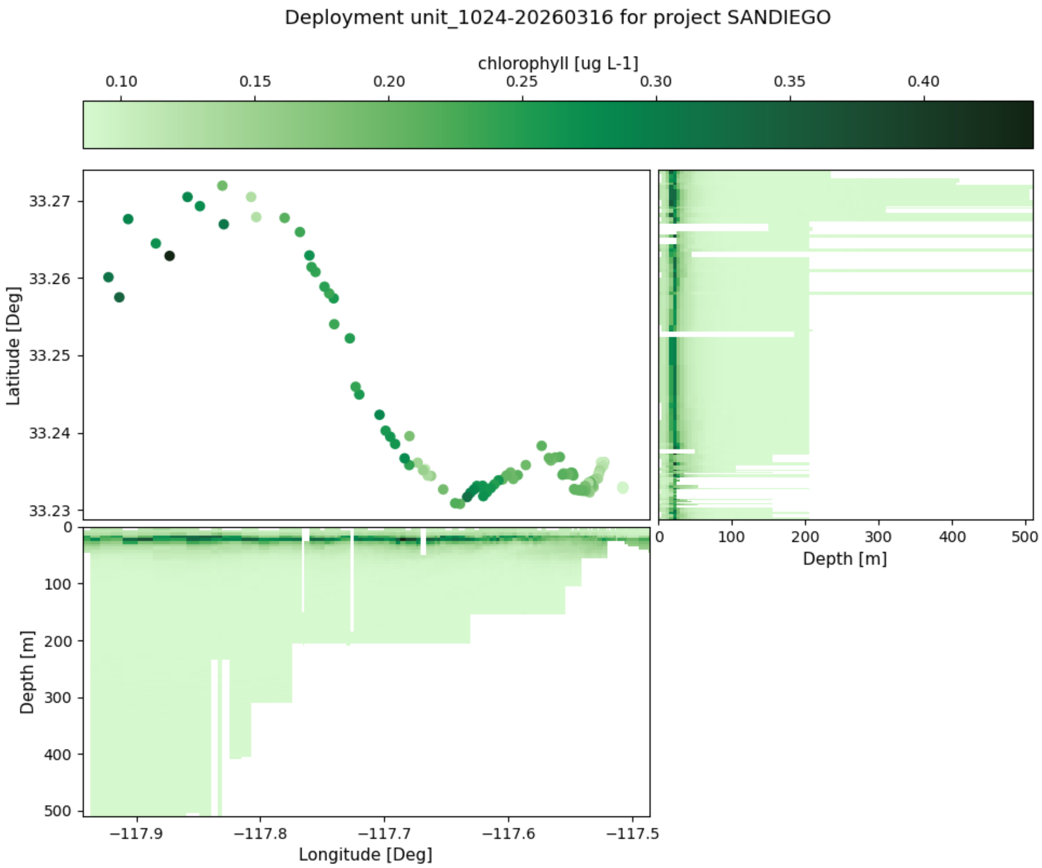
Once the glider was back in the laboratory, pilots inspected and cleaned the glider. All data were offloaded from the science and flight cards, and photos were offloaded from the camera. All files were uploaded to NOAA's Google Cloud Platform (see Data Management structure and access to data here). Data will be quality checked and analyzed at a later date.

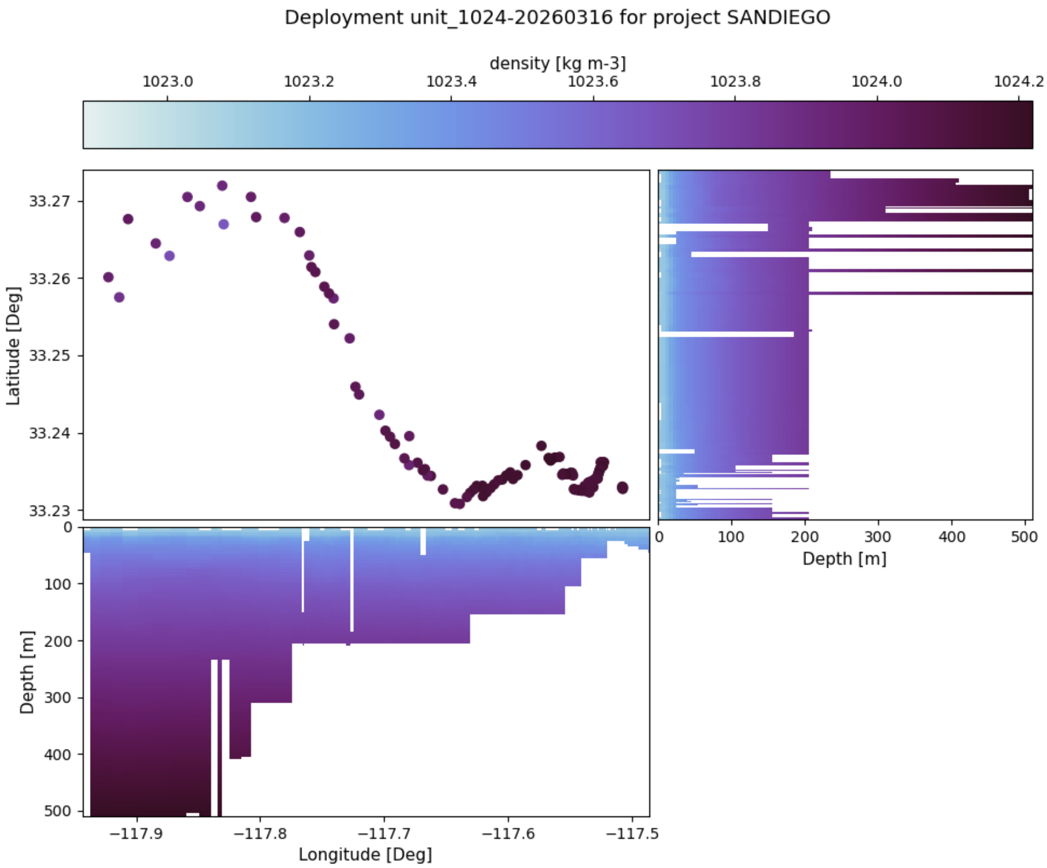
Figures

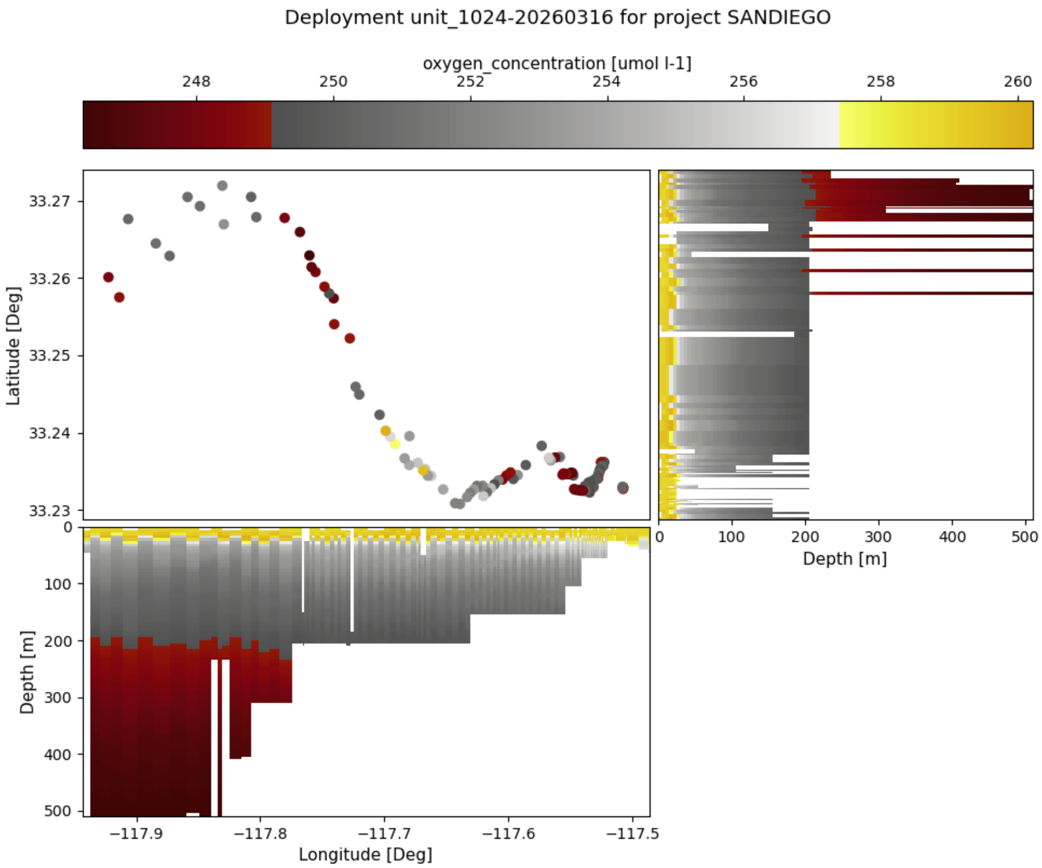
Plots below are generated from raw data which has not yet been quality-checked.



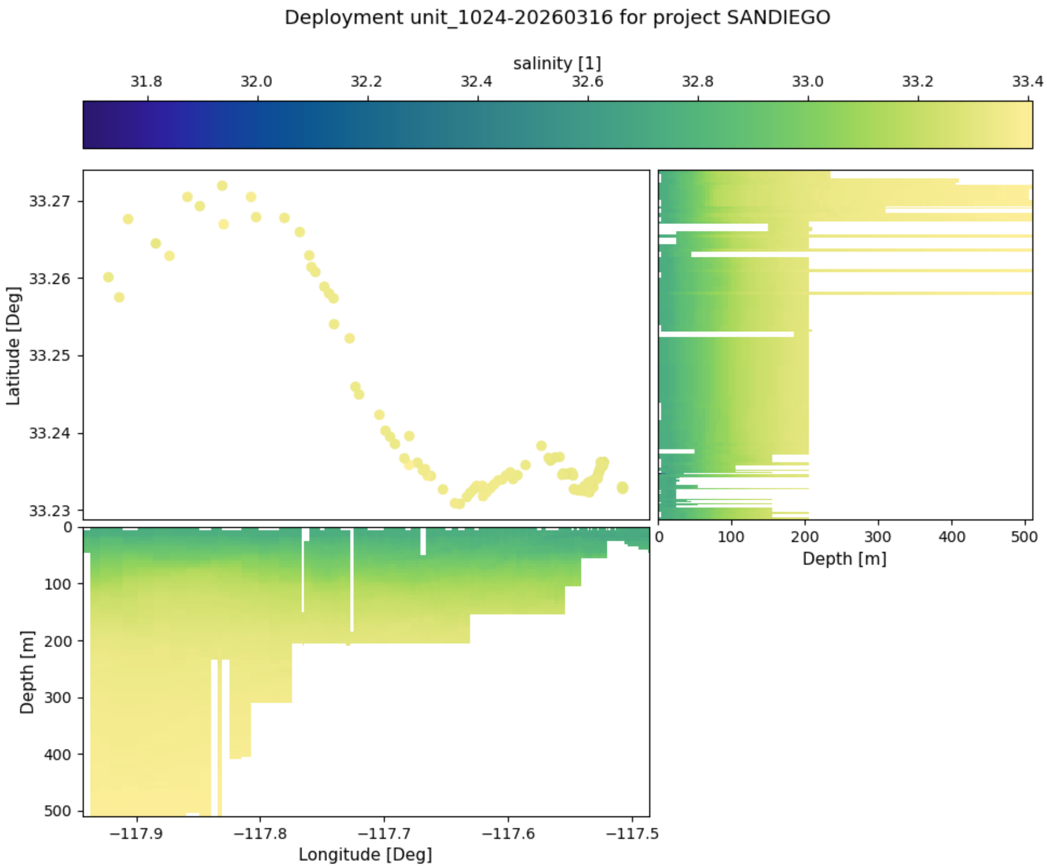
Deployment unit_1024-20260316 for project SANDIEGO
backscatter_700 [1]
0.00020
0.00025
0.00030
0.00035
0.00040
33.27
33.26
33.25
33.24
33.23
Latitude [Deg]
0
100
200
300
400
500
Depth [m]
−117.9
−117.8
−117.7
−117.6
−117.5
Longitude [Deg]

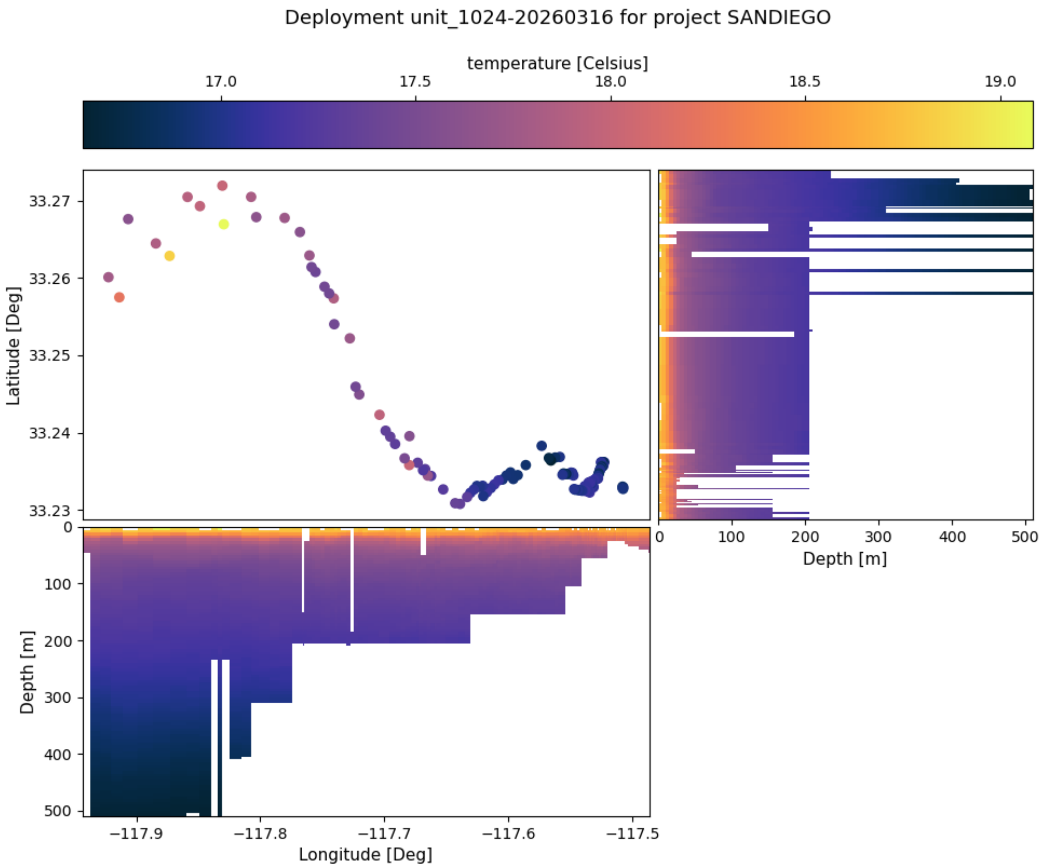

Deployment unit_1024-20260316 for project SANDIEGO
chlorophyll [ug L-1]
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Latitude [Deg]
33.27
33.26
33.25
33.24
33.23
Depth [m]
0
100
200
300
400
500
Longitude [Deg]
−117.9
−117.8
−117.7
−117.6
−117.5
Depth [m]
0
100
200
300
400
500


Deployment unit_1024-20260316 for project SANDIEGO
density [kg m-3]
1023.0
1023.2
1023.4
1023.6
1023.8
1024.0
1024.2
Latitude [Deg]
33.27
33.26
33.25
33.24
33.23
Depth [m]
0
100
200
300
400
500
Longitude [Deg]
−117.9
−117.8
−117.7
−117.6
−117.5



Deployment unit_1024-20260316 for project SANDIEGO
oxygen_concentration [umol l-1]
248
250
252
254
256
258
260
33.27
33.26
33.25
33.24
33.23
Latitude [Deg]
0
100
200
300
400
500
Depth [m]
0
100
200
300
400
500
Depth [m]
−117.9
−117.8
−117.7
−117.6
−117.5
Longitude [Deg]


Deployment unit_1024-20260316 for project SANDIEGO
salinity [1]
31.8
32.0
32.2
32.4
32.6
32.8
33.0
33.2
33.4
33.27
33.26
33.25
33.24
33.23
Latitude [Deg]
0
100
200
300
400
500
Depth [m]
0
100
200
300
400
500
Depth [m]
−117.9
−117.8
−117.7
−117.6
−117.5
Longitude [Deg]



Deployment unit_1024-20260316 for project SANDIEGO
temperature [Celsius]
17.0
17.5
18.0
18.5
19.0
Latitude [Deg]
33.27
33.26
33.25
33.24
33.23
Depth [m]
0
100
200
300
400
500
Depth [m]
0
100
200
300
400
500
Longitude [Deg]
−117.9
−117.8
−117.7
−117.6
−117.5